What Feeds the Acuity Number

Inputs, Weights, and Why Each One Is There

Every characteristic used by `predict_acuity()`, its numerical weight in the final result, and the source or reasoning behind it

Traced against a live run: 400 µm, $M = 8$ D, verified 2026-09-09

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Chapter 1

The final equation, and what each piece contributes

1.1 The formula

$$\mathrm{MAR}_{\mathrm{arcmin}} = \underbrace{\frac{30}{f_{\mathrm{MTF10}}}}_{\text{(1) optical cutoff}} \times \underbrace{\mathrm{VSOTF}^{-0.1}}_{\text{(2) neural weighting}}, \qquad \mathrm{logMAR} = \log_{10}(\mathrm{MAR}) + \underbrace{\Delta_{\mathrm{bright}}}_{\text{(3) illuminance}} \tag{1.1}$$

Three characteristics of the optical system feed this equation. Nothing else does. This chapter quantifies exactly how much each one moves the final number, using a live traced run at 400 µm, $M = 8$ D as the worked example throughout — every number below was read directly from a single call to `run_stage4_point()`, not estimated.

Characteristic	What it measures	Weight in final logMAR	At this operating point
(1) MTF10 cutoff frequency	Optical resolving power of the pinhole+eye system	**dominant**	14.44 cpd → logMAR +0.3175
(2) VSOTF (neural-weighted Strehl)	How much of that optical signal survives visual processing	**small**	0.739 → shifts logMAR by +0.0131
(3) Retinal illuminance	Whether the image is bright enough for full acuity	**usually zero**	1257 Td, above the 300 Td knee → shifts logMAR by 0.0000

Table 1.1: The three inputs, ranked by actual effect size at the design's own operating point.

The one-sentence summary

Acuity is set almost entirely by where the modulation transfer function crosses 10% contrast. The neural weighting nudges it by about a hundredth of a logMAR unit. The brightness term is fully wired in, cited, and currently contributes nothing to any of the eight fabricated designs — because every one of them, at the assumed scene luminance, sits above the threshold where it would matter.

Chapter 2

External validation status — read this before quoting any predicted logMAR

The predictor has not been validated against real clinical data, and the one attempt failed

The full model (MTF10 + VSOTF + brightness correction, exactly as traced in this document) was run against Wang et al. (2009)'s real clinical pinhole-acuity measurements, in `stage4_human_vision/RESULTS.md`.

M (D)	R^2	Pearson r	p-value
0	−6.39	0.395	0.510
2	−4.98	0.393	0.513
5	−2.84	0.405	0.499

Negative R^2 means the model predicts worse than simply guessing the mean every time. $p \approx 0.50$ means no statistically significant relationship was detected. This is not weak correlation — every one of the 15 tested (aperture, M) points shows the *same-direction* bias: the model over-predicts blur (worse logMAR than observed) at every point tested.

The luminance assumption was tested and ruled out as the cause. Correcting the scene luminance from the 10 000 cd/m^2 default to Wang 2009's own reported illumination (15 footcandles → 43.7 cd/m^2) made R^2 *worse* (−7.88 vs. −6.68 at $M = 0$), because it pushed apertures further below the 300 Td knee and added predicted blur on top of a model that was already over-predicting it (`LUMINANCE_RECONCILIATION.md`).

The project's own leading diagnosis, most likely cause first: (1) the acuity formula has no fitted offset or scale against real data — `DESIGN.md` flags this as a known "requires empirical calibration" gap; (2) the Wang 2009 dataset is non-monotonic at $n = 5$ (observed logMAR 0.28, 0.31, **1.00**, 0.40, 0.50 at 500/750/1000/1200/1500 µm — a spike at 1000 µm that reverses at 1200) and a single outlier at $n = 5$ sinks any smooth model regardless of which optical metric underlies it; (3) the dataset records no per-subject refractive error, adding irreducible noise; (4) $n = 5$ is too small for significance even with real signal present. `RESULTS.md`'s own conclusion: this "cannot be properly validated with currently available public data."

How to use every number in this document, given the above

Every logMAR value quoted here is a **self-consistent model output** — traceable, cited term by term, internally coherent — *not* a validated clinical prediction. Say the validation status yourself if the acuity model comes up: internally consistent, externally unvalidated, the one validation attempt failed, luminance was tested and ruled out as the cause, the leading suspects are lack of calibration and a small non-monotonic clinical dataset rather than the underlying optics. This is also the reason the project's own logMAR measurement protocol matters more than a confirmatory afterthought — it is calibration data this model currently has none of.

Chapter 3

Characteristic 1: MTF10 cutoff frequency — the dominant term

3.1 What it is

The spatial frequency, in cycles per degree, at which the eye's modulation transfer function falls to 10% contrast. Below that frequency, detail is still transmitted with enough contrast to see; above it, the eye cannot reliably tell light from dark at that fine a spacing.

3.2 Where it comes from

Computed in `optical_quality.py` from the retinal point-spread function that Stage 2's wave propagation produces. The PSF is radially averaged into an MTF via `stage3_metrics.mtf_radial`, then the frequency at 10% is read off by `mtf50_mtf10`. This is a direct measurement of the simulated optical system — not a fitted or assumed number.

3.3 How it enters the acuity formula

$$\mathrm{MAR}_{10} = \frac{30}{f_{\mathrm{MTF10}}\,[\mathrm{cpd}]} \tag{3.1}$$

The constant 30 is `cycles_per_degree_at_20_20` — the standard clinical definition that 20/20 (logMAR 0) acuity corresponds to resolving 30 cycles per degree. This is the entire definitional anchor of the whole model: everything downstream is expressed relative to this one reference number.

Worked example, 400 µm at $M = 8$ D: MTF10 = 14.4431 cpd, so $\mathrm{MAR}_{10} = 30/14.4431 = 2.0771$ arcmin, giving $\log_{10}(2.0771) = +0.3175$ logMAR before any other correction is applied. This is 96% of the final answer (+0.3306).

Why this is the dominant term by design, not by accident

The model's philosophy is: acuity is fundamentally an optical-resolution question, and everything else (neural weighting, illuminance) is a secondary correction on top of that resolution limit. That is a defensible simplification — MTF10 alone is a standard, clinically-used proxy for acuity in its own right (it is how many autorefractors and wavefront-aberrometer-based acuity predictions work) — but it does mean the model is only as good as the PSF that produced this one number. Every validation concern raised elsewhere in this project about the retinal PSF (the FWHM ring-hopping, the 8% Airy tolerance at the first null) is inherited directly into this term, because it is computed from the same PSF.

Chapter 4

Characteristic 2: VSOTF — neural-weighted optical quality

4.1 What it is

Visual Strehl ratio computed from the OTF, or VSOTF: a single number summarising how much of the optical signal survives once the eye's own contrast sensitivity is factored in, normalised against the diffraction-limited case.

4.2 Where it comes from

$$\text{VSOTF} = \frac{\int \text{OTF}_{\text{real}}(f)\,\text{CSF}(f)\,df}{\int \text{OTF}_{\text{real,diffraction-limited}}(f)\,\text{CSF}(f)\,df} \tag{4.1}$$

computed in `visual_strehl.py`. The numerator integrates the *actual* optical transfer function — weighted by the eye's contrast sensitivity function (CSF), so frequencies the eye barely perceives contribute little to the integral. The denominator is the same integral for a perfect, diffraction-limited system at the same aperture. VSOTF = 1 means the system performs as well as physically possible; lower values mean real aberrations (here, defocus) are costing performance.

The CSF weighting uses the **Barten model** by default (`csf.py`, `barten_simplified_csf`), a luminance- and field-size-dependent closed form. Two alternative CSF models (Mannos–Sakrison, a normalised Watson template) are computed alongside it for comparison but are not used in the recommended acuity number.

4.3 How it enters the acuity formula

$$\text{MAR}_{\text{adjusted}} = \frac{\text{MAR}_{10}}{\text{clamp}(\text{VSOTF},\,0.25,\,2.0)^{0.1}} \tag{4.2}$$

Worked example: VSOTF = 0.7389. $0.7389^{0.1} = 0.9700$. $\text{MAR}_{\text{adjusted}} = 2.0771/0.9700 = 2.1409$ arcmin, moving logMAR from +0.3175 to +0.3306 — a shift of **0.0131 logMAR**.

Why the exponent is 0.1, and why that makes this a deliberately weak correction

The clamp restricts VSOTF to $[0.25, 2.0]$ before the exponent is applied, and the exponent itself is small. Even at the clamp's lower bound, $0.25^{-0.1} = 1.149$, a MAR increase of only 15% (≈ 0.06 logMAR). This is a design choice: VSOTF is being used as a *fine correction* to the MTF10-based estimate, not as an independent, comparably-weighted acuity predictor in its own right. The reasoning is that MTF10 already captures most of what the degraded optics do to acuity; VSOTF adds the additional information that some of the remaining contrast is more or less usable to the visual system depending on which spatial frequencies survived, without being allowed to swing the answer by more than a fraction of a line.

4.4 What the Barten CSF weighting itself depends on

Two further characteristics enter here, both fixed defaults rather than swept design variables in your current results:

Input	Default	Effect
Scene luminance	$10\,000\,\mathrm{cd/m^2}$	Higher luminance raises peak CSF sensitivity and shifts the CSF's high-frequency cutoff outward (via the a and b terms in `barten_simplified_csf`) — brighter scenes let the eye use finer detail, so VSOTF's frequency weighting shifts toward higher spatial frequencies at high luminance.
Field diameter	20°	Enters the same a term; a larger field lowers the $12/w$ term in the denominator, slightly raising sensitivity. This models spatial summation across a wider retinal area.

Chapter 5

Characteristic 3: retinal illuminance — present, cited, usually zero

5.1 What it is

The amount of light actually reaching the retina, in trolands (Td) — a unit that already accounts for pupil area, so it is a direct measure of retinal light flux rather than scene brightness alone.

5.2 Where it comes from

Two steps, in `illuminance.py` and `pupil.py`:

Step A — the natural pupil diameter. The Watson–Yellott (2012) unified pupil model, solved iteratively because pupil size and retinal illuminance are mutually dependent (illuminance depends on the effective stop, which depends on pupil size, which depends on illuminance). Converges in a handful of iterations to 0.0001 mm tolerance.

Step B — the effective stop. Whichever is smaller, the pinhole or the natural pupil:

$$d_{\text{stop}} = \min(d_{\text{pupil}},\, d_{\text{pinhole}}) \tag{5.1}$$

For every one of your fabricated designs the pinhole is smaller (e.g. 0.400 mm vs. a natural pupil of 3.725 mm at $M = 8$), so **the pinhole is the effective aperture stop in every case** — this is the whole physical point of the device.

$$\text{Td} = L_{\text{scene}}\,[\text{cd/m}^2] \times A_{\text{stop}}\,[\text{mm}^2] \tag{5.2}$$

5.3 How it enters the acuity formula

$$\Delta_{\text{bright}} = \begin{cases} 0 & \text{Td} \geq 300 \\ 0.060 \times \log_{10}(300/\text{Td}) & \text{Td} < 300 \end{cases} \tag{5.3}$$

The 300 Td knee is from Van Nes & Bouman (1967): above it, contrast detection is governed by the Weber–Fechner law and is illuminance-independent; below it, the de Vries–Rose regime applies and sensitivity falls off. The 0.060 logMAR-per-decade slope is from Tidbury, Czanner & Newsham (2016), verified against their Table 3 ($p < 0.001$); the code comments record that an independently-derived alternative slope from Sheedy, Bailey & Raasch (1984), ≈ 0.0664, could only be traced to a secondary summary and so was not used.

Worked example, 400 µm at $M = 8$ D: Td = 1256.6, above the 300 Td knee, so $\Delta_{\text{bright}} = 0$.

The finding: this term is inert for every fabricated design, and you should say so first

Aperture (µm)	Retinal illuminance (Td)	Brightness penalty (logMAR)
75	44.2	+0.0499
100	78.5	+0.0349
150	176.7	+0.0138
200	**314.2**	+0.0000
	every fabricated design (400–2000 µm) falls here	
2000	31 415.9	+0.0000

The knee sits at 200 µm. Every one of your eight gaze-weighted winners is at 400 µm or larger. At the model's assumed 10 000 cd/m^2 scene luminance, **the brightness term contributes exactly zero logMAR to every design you have actually selected.** It is real physics, correctly cited, correctly implemented — and currently silent. It would activate for an indoor measurement (typical luminance 100–300 cd/m^2, roughly two decades below the assumption) or for any future aperture below 200 µm. State this yourself: it pre-empts the question of whether you actually tested the brightness term's effect on your headline result, and the honest answer is that you have not needed to, because it does not act on that result.

Chapter 6

What is computed but does NOT enter the recommended number

Two further quantities are computed by the pipeline and reported alongside the recommended acuity value, but neither contributes to it. Listing them matters because they appear in the same output dictionary and could be mistaken for inputs if read from the code without care.

6.1 MTF50

The 50%-contrast cutoff frequency, computed identically to MTF10 but at a higher contrast threshold. Used only to bound the reported `approximate_ci_95_mar_arcmin` interval — which, despite the name, is **not a statistical confidence interval**. It is [min, max] of three different model choices (MTF10-based, MTF50-based, and the recommended VSOTF-and-brightness-adjusted value), not a distribution with a defined confidence level.

Do not present this interval as a confidence interval

At $M = 8\,\mathrm{D}$ it spans logMAR +0.318 to +0.795 — 4.8 chart lines. If shown, label it "spread across model choices," not "95% CI." An examiner asking what distribution produced a 95% figure has no defensible answer, because none was computed.

6.2 The legacy SNR term

An older acuity model, retained in the code as `vsotf_snr_legacy` for direct comparison rather than deleted. Uses an uncited $\mathrm{SNR}^{0.05}$ exponent in place of the cited Tidbury brightness correction. The project's own change log records that this was superseded specifically because it lacked a literature source; it remains computable but is not the recommended value and should not appear as a headline number in the presentation.

Chapter 7

Full input inventory

Every value that can change the output of `predict_acuity()`, traced to its origin.

Table 7.1: Complete input inventory for the acuity prediction.

Input	Source	Design variable?	Reasoning / citation
Aperture diameter a_{hole}	user-set	**yes — swept**	The physical design parameter under optimisation
Wavelength	user-set, default 550 nm	fixed	Photopic luminosity peak
Gap distance d_{gap}	user-set, default 12 mm	fixed	Spectacle vertex distance
Myopia M	user-set	**yes — swept**	The physiological variable under study
P_0, n'	60 D, 1.336	fixed	Standard emmetropic eye power and vitreous index
Scene luminance	10 000 cd/m^2	fixed default	Bright daylight reference; see Ch. 3 warning on indoor measurement
Field diameter	20°	fixed default	Typical CSF measurement field size
Subject age	28.58 yr	fixed default	Watson–Yellott reference age
Eyes (mono/bino)	2 (binocular)	fixed default	Affects pupil-model flux density only
CSF model	Barten	fixed default	Most widely validated luminance-dependent CSF; two alternatives computed but unused
Cycles/deg at 20/20	30.0	fixed constant	Clinical definition, the entire scale anchor
VSOTF exponent	0.1	fixed constant	Deliberately weak; see Ch. 2 keybox
VSOTF clamp	$[0.25, 2.0]$	fixed constant	Bounds the correction's maximum possible swing
Weber knee	300 Td	fixed constant, cited	Van Nes & Bouman (1967), verified primary source
Brightness slope	0.060 logMAR/decade	fixed constant, cited	Tidbury et al. (2016), verified Table 3

The reasoning behind the split between swept and fixed

Only aperture and myopia vary across your actual results — every other quantity in this table is a fixed default representing a specific, stated measurement scenario (bright outdoor use, average adult pupil dynamics, standard CSF assumptions). This is a reasonable scoping choice for a design optimisation, but it means the predicted numbers are conditional on that scenario holding. Chapter 3's finding — that the brightness term is inert at this luminance — is a direct consequence of that choice, and is the one fixed default most likely to matter if the testing conditions differ from the model's assumption.